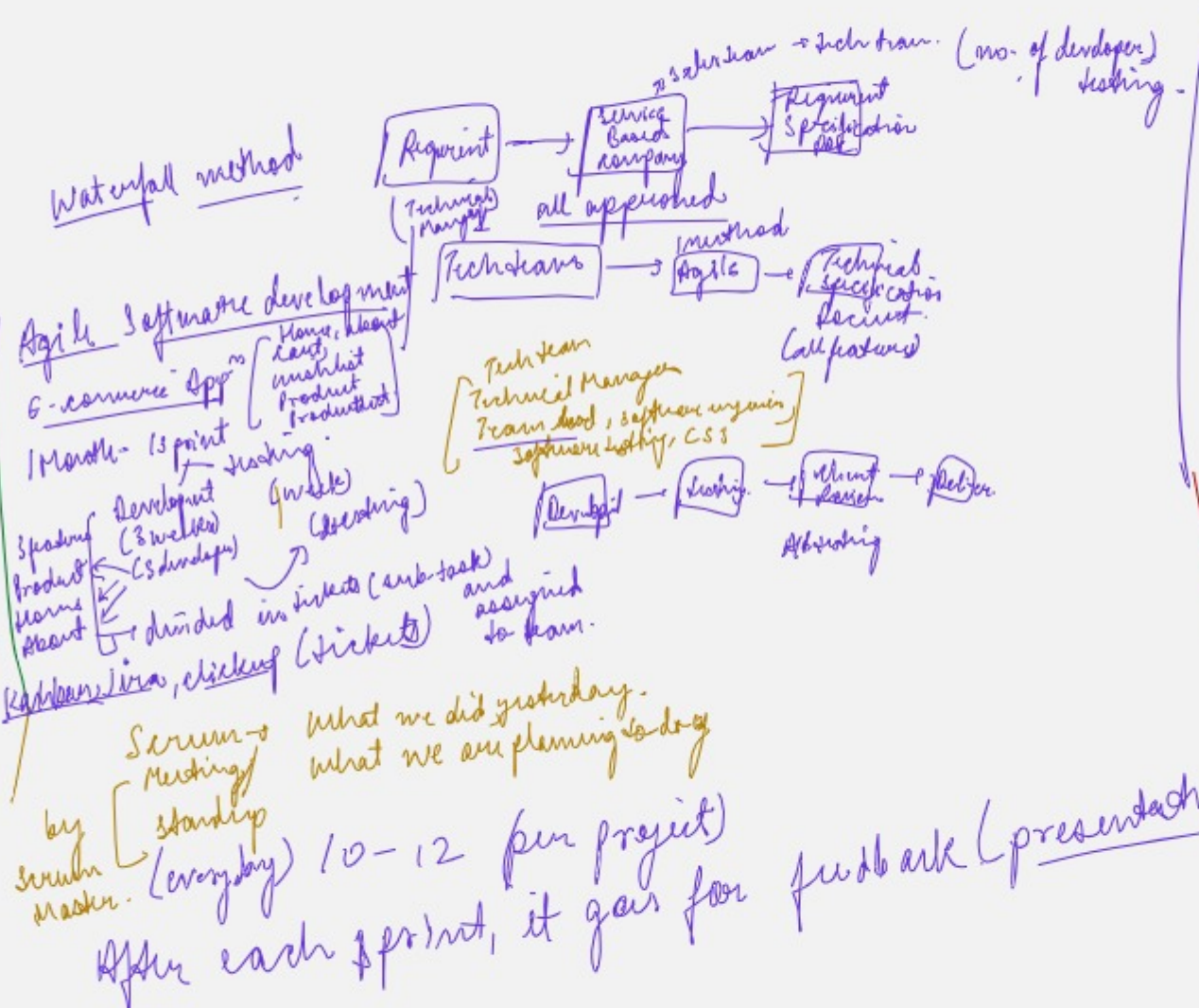
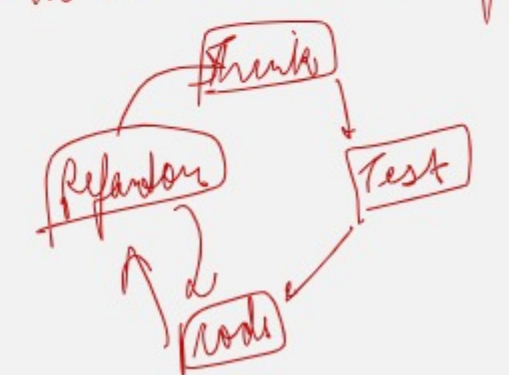


- Scalability
↳ Bottlenecks that hurt the scalability
1. Poor code quality
 2. Choosing the wrong DB type
 3. Not caching
 4. Load balancer
 5. Wrong API (type)
 6. Handling sync or async



TDD (Google, Facebook)

- ↳ Test driven development
- ↳ write test case before development.



CI/CD (Workflow)

Continuous Integration & Continuous Delivery

- ↳ code changes
- ↳ code integration
- ↳ build and test
- ↳ rapid feedback

Automated Deployment pipeline

Deployment to staging (env.)

Automated release to Prod.

1. efficiency improves
2. faster the deployment
3. quality assurance and bug detection
4. collaboration
5. risk reduction

Dev Sec Ops

Development, Security, Operations

Development (Dev)

- ↳ CI
- ↳ Automated testing
- ↳ code review

Security (Sec)

- ↳ SAST - static application security testing
- ↳ DAST - security testing in deployment env.
- ↳ security assessment

Ops

- ↳ ICD
- ↳ PaE (Infrastructure as code)
- ↳ Monitoring and incident response

Technical Team

Performance Optimization

(Website)

1. code optimization (unnecessary loop, memory allocation)
2. caching
3. database optimization
4. network optimization
5. compressed and Minification of images

6. CDN
7. Parallelization
8. process multitask parallelly (asyn. prog.)